Application: Evaluating food access strategies in Austin to improve healthy food consump

Joy Casnovsky - joy@sustainablefoodcenter.org
Tipping Points

Summary

ID: CA18-SS-0000000050
Last submitted: Dec 17 2021 06:02 PM (EST)

Final Scientific Progress Report

Completed - Dec 17 2021

Form for "FFAR Annual Progress Report"

Grant Information*

Award Program	Tipping Points
Grant ID	CA18-SS-0000000050
Project Title	Evaluating food access strategies in Austin to improve healthy food consumption and food security
Grant Start Date	2/1/2018
Grant End Date	9/30/2021
Total Budget	$2,197,581.62
Matching Funder(s)	City of Austin
Final Report Date	12/15/21

Project Director/Principal Investigator Information*

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Grantee Organization Information

Organization Name	Sustainable Food Center, Inc.
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City	Austin
State	Texas
ZIP	78702
EIN	74-2441468

Authorized Signing Official Information

Name	Mr. Mark Bethell
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1. GENERAL INFORMATION

Notes on Spending

If you've experienced any deviation in the spending for your award for the last reporting period such as changes to the amount of source or matching funds, changes from your original spending plan, or carry-forward, please provide a justification below. You may enter "N/A" or "Nothing to report" if you did not experience any special circumstances.

Approximately $12,839.28 was left in the UTHealth Subcontract due to: 1) their subcontractor did not invoice for all allocated funds and 2) funds remain due to virtual conferences as opposed to in-person conferences.

1.1. Please list the geographic location(s) – city, state, congressional district - where the work was conducted. If the work was conducted outside of the U.S., please list the city and country.*

Austin, TX. Congressional districts: 10, 17, 21, 35.

1.2. How many new jobs were created by the grant during this reporting period? *

During June 2020-May 2021, there were four new jobs created.

1.3. How many jobs were maintained by the grant during this reporting period?*

Six

1.4. Have there been any changes to your organization's IRS 501(c)(3) non-profit status since you were awarded the grant?*

No

If yes, please explain:

n/a

1.5. Has your organization undergone a recent name change? If so, please provide the new name of your organization.*

No

2. SCIENTIFIC REPORT

2.1. Public Abstract*

The abstract should be 'stand-alone' and is intended for a general audience. It should be a concise overview/summary of the importance of the project, the issues that the project addressed, and the key findings of the project. The abstract should also clearly state how the results of the project help to address important needs in U.S. food and agriculture systems.

Introduction: The absence of equitable and accessible food retail is a contributor to food insecurity and limits the ability of residents of low-income communities to consume healthy diets. In 2018, the city of Austin started the implementation of the Fresh for Less initiative, which uses innovative strategies (i.e. farm stands, mobile markets, healthy corner stores and financial incentives) to increase healthy food access in low-income communities. We designed an evaluation study within the context of this natural experiment to assess the impact of the Fresh for Less initiative on food insecurity and fresh fruit and vegetable (FV) purchase/consumption among low-income, diverse communities, and inform an agent based model (ABM) for predicting changes in vegetable intake under different policy expansion scenarios (Study 1). At the start of COVID-19, our research team added a second study to examine the short- and longer-term impacts of COVID-19 on food insecurity, food acquisition, and consumption (Study 2).

Methods: 400 participants were recruited from low-income communities and categorized into 3 groups: (1) Confirmed Users: participants recruited at Fresh for Less locations (n=130), (2) Geographically Exposed: participants living within 1.5 mile of a Fresh for Less location (n=185), and (3) Comparison: participants living in communities with no Fresh for Less locations (n=85). Participants completed a survey during fall 2018 and 2019. The survey measured demographics and outcomes: food insecurity and fresh FV purchase and consumption. Other measures included accelerometers, GPS devices, store-

level audits, and built environment assessments. Baseline data were used to develop and calibrate the ABM. For Study 2, original cohort participants completed COVID-19 Surveys in April/May and Nov/Dec, 2020.

Results: At baseline, participants were 54% Hispanic and 40% food insecure. Food insecurity decreased significantly among Geographically Exposed and Confirmed Users between baseline and year 2. ABM findings indicate that increasing the number of mobile markets and farm stand locations is effective for increasing vegetable consumption. When combining an increase in number of locations with a 50% discount on vegetables, greater increases in vegetable intake were predicted. However, vegetable consumption did not increase meaningfully when increasing geographic or economic access in healthy corner stores. Offering discounts greater than 50% for vegetables at traditional food stores, like supermarkets, was predicted to result in similar gains in vegetable intake as observed when adding 30 more mobile markets without discounts, or 15 more mobile markets with a 50% discount. Study 2 showed greater persistence of food insecurity during COVID-19 among Hispanic/Black participants compared to white participants. Food assistance programs were utilized regardless of food insecurity status, but among consistently food insecure respondents, food bank usage was higher, while FV purchasing decreased.

Conclusions: Food insecurity disparities have widened during COVID-19. Our results provide important insights for prioritizing geographic and economic access strategies to increase FV intake and food security among low-income communities in Central Texas. Jointly increasing geographic and economic access to mobile markets, and/or offering discounts for FV purchases in supermarkets can be synergistic. Policies for improving access and food security in low-income communities can also benefit local farmers.

2.2 Provide a description or interpretation of how the key findings of the project can be used by the agriculture community. It is important the PI identifies how these findings have been, or may be, adopted or adapted in U.S. agriculture and food systems. If the findings cannot yet be applied, this section should address how they can be used to guide future planning or decision-making.*

Our study used a combination of empirical (quantitative and qualitative) and simulation approaches to understand how geographic and economic strategies to increase access to fresh fruits and vegetables influence food-related behaviors among low-income, urban Central Texas communities. Our key findings were as follows:
1. A significant proportion of the studied population experiences food insecurity (40%), and this is higher

among racial/ethnic minorities (Hispanics and Blacks), underscoring systemic inequities in the food system that impact population health and wellbeing.

2. COVID-19 exacerbated pre-existing inequalities in access to healthy food among low-income, diverse communities.

3. Innovative strategies aiming to increase geographic access to healthy foods, such as the Fresh for Less initiative, which increases access to fresh fruits and vegetables in low-income areas via the strategic placement of mobile markets, farm stands, and healthy corner stores, should consider the cost-effectiveness of each approach before large-scale investment and expansion. Our simulation results suggest that:

a. Investing in expanding access to fresh fruits and vegetables via healthy corner stores will not significantly modify consumption of vegetables among low-income, urban residents.

b. Among the three types of strategies included in Fresh for Less, mobile markets show the most promise for positively improving access to and consumption of healthy foods among low income residents. Investing in expanding the density (number of stores) of these types of stores in low-income neighborhoods can be an effective strategy for improving vegetable intake, and in consequence, may help reduce food insecurity. Optimal benefits are predicted if this type of approach is implemented in concert with a meaningful (50% or more) discount in the cost of vegetables.

c. Investing in building partnerships with traditional food vending outlets, like supermarkets and small grocers, as assets where economic incentives for healthy eating can be implemented, can be as good of a strategy for improving healthy eating and food security in low-income communities, as is investing in other innovative, yet less traditional approaches (like Fresh for Less). Specifically, offering significant discounts in the cost of vegetables at these stores (50% or more), which could be exclusive to low-income residents, could be a very effective strategy for improving access to healthy food and food security. The highest expected benefits occur when the discount offer is of 75% or more.

Our findings are of critical relevance for achieving equitable, sustainable, and healthy agricultural and food systems in the US, as this requires careful consideration of the multiple levers of influence on the food system, from food production and agriculture, to distribution, to consumption by the public. Our work provides important insights on the complexities of the food system within the realm of food access in stores by low-income, diverse urban consumers.

Our results suggest that policies supporting subsidies for local farmers to stock supermarkets, small grocers, and mobile markets in low-income communities, whereby these can be offered at significantly reduced cost to the consumer, can offer synergistic benefits on the production and consumption end, whilst maximizing efforts to secure access to healthy food for all. Likewise, increasing the density of mobile markets, and implementing regulations that ensure that local farmers can distribute and sell their products directly to the consumer through these outlets, could offer an opportunity for savings in transportation and storage costs for farmers, whilst maximizing geographic access to healthy food offerings for low-income residents.

Notably, our findings represent only the first set of simulated scenarios for which we have employed our agent-based model for. These types of methods remain underutilized as a means of providing evidence-based support for large-scale improvements to food systems.

The model that we developed represents a powerful tool to pre-test possible strategies for modifying the geographic and economic landscape of the food system in Austin, Texas, before committing to investing in their implementation and scale up. This type of bi-directional exercise, whereby what the model tests becomes informed by the priorities and ideas of key stakeholders in the food system, while researchers provide findings that can inform policy action, can be very helpful in determining which strategies are more likely to yield successful and cost-beneficial outcomes, and allows to re-examine priorities in a timely fashion as more empirical date becomes available.

2.3. Research Outcomes/Impact *

The primary goal here is to answer questions such as “How did this project lead to improvements in U.S. agriculture and food systems?” or “How can the findings of this study guide or make improvement in future investigations and research?” Outcomes should be explained and classified in one of the following ways:

- Potential outcomes, i.e., findings, results, or recommendations that could impact U.S. agriculture and food systems, if used;
- Intermediate outcomes, i.e., how the findings, results, or recommendations have been used by others to influence practices, legislation, research/product design, training and so forth; and
- End outcomes, i.e., how findings, results, or recommendations have contributed to documented reductions in work-related morbidity, mortality, and/or exposure.

Several manuscripts presenting results from this study have been published. Please refer to those manuscripts for detailed information about results. In this section of the report we are just highlighting some key findings.

In Table 1, the reader will find sociodemographic characteristics of our participants according to recruitment strategy at baseline. In Table 2, we present Fruit and vegetable purchasing, food shopping behaviors and motivations among adults at baseline. These data were used as baseline data for our evaluation study but also as data for the Agent-based Model we created.

Baseline results show that the three groups of participants were generally comparable; however, there were statistically significant differences among some sociodemographic characteristics and food consumption as well as shopping behaviors. Specifically, there were statistically significant differences by various sociodemographic factors including income, education, food assistance utilization, and food insecurity status. For the aforementioned factors, there were the most notable differences among Confirmed Users of FFL. This demonstrates that the FFL program is reaching its intended communities of interest, since they were designed to improve access to healthy foods among individuals with lower income and socioeconomic status and those experiencing food insecurity. Moreover, there were significant differences among various behavioral variables of interest including fruit consumption and shopping behaviors, specifically utilization of different types of food retail and FFL utilization.

Table 3 presents changes in some of the food acquisition and consumption behaviors from baseline to year 2 according to recruitment group. Results indicate a significant increase in shopping at a Mobile Market among the Confirmed Users group. Food insecurity decreased significantly in both the Confirmed Users and the geographically Exposed groups. In addition, the Geographically Exposed group reported a significant increase in fruit consumption.

Lastly, Table 4 presents data from Study 2 which shows sociodemographic characteristics of our sample by food security transition status.

Results from simulations:

4. Our simulation results (See Figure 1) suggest that:

a. Investing in expanding access to fresh fruits and vegetables via healthy corner stores will not significantly modify consumption of vegetables among low-income, urban residents.

b. Among the three types of strategies included in Fresh for Less, mobile markets show the most promise for positively improving access to and consumption of healthy foods among low income residents. Investing in expanding the density (number of stores) of these types of stores in low-income neighborhoods can be an effective strategy for improving vegetable intake, and in consequence, may help reduce food insecurity. Optimal benefits are predicted if this type of approach is implemented in concert with a meaningful (50% or more) discount in the cost of vegetables.

c. Investing in building partnerships with traditional food vending outlets, like supermarkets and small grocers, as assets where economic incentives for healthy eating can be implemented, can be as good of a strategy for improving healthy eating and food security in low-income communities, as is investing in other innovative, yet less traditional approaches (like Fresh for Less). Specifically, offering significant discounts in the cost of vegetables at these stores (50% or more), which could be exclusive to low-income residents, could be a very effective strategy for improving access to healthy food and food security. The highest expected benefits occur when the discount offer is of 75% or more.

(See also, attached:
- Figure 1. Increasing Locations and Financial Discounts
- Table 1. Descriptive statistics of sociodemographic characteristics stratified by recruitment strategy among adults at baseline (FRESH Evaluation study Austin, TX, USA, October 2018–March 2019).
- Table 2. Fruit and vegetable purchasing, food shopping behaviors and motivations among adults at baseline (FRESH Evaluation study Austin, Texas, (October 2018–March 2019).
- Table 3: Changes baseline – year 2 adjusted for Race/Ethnicity and Income
- Table 4. Demographic characteristics of the sample stratified by food insecurity transition status.)

2.4. What were the goals/specific aims of the project? If the approved application lists milestones/target dates for important activities or phases for this reporting period, identify these milestones and dates, as well as show actual completion dates or the percentage of completion of milestone targets.*

All goals and milestones during the third year of the study have been reached. Completed goals of the project during the reporting period (June 2020 – December 2021):

1. Received IRB approval for modifying scope of study, new protocols and survey tools to account for COVID-19 specific analyses.
2. Created and administered two COVID-19 surveys in June/July 2020 and November/December 2020 to 367 and 304 survey respondents, respectively.
3. Conducted over-the-phone qualitative interviews with 82 cohort participants.
4. Presented two oral papers to the American Public Health Association's (APHA) Annual Meeting in October 2020.
5. Presented one oral paper to the APHA's Annual Meeting in October 2021.
6. Presented on two separate panels at the Conference on Food Resilience and Equity in January 2021 and in November 2021.
7. Submitted four abstracts to the International Society of Behavioral Nutrition and Physical Activity Annual Meeting in May 2022
8. Presented to the City of Austin and Episcopal Health Foundation.
9. Presented two webinars hosted by the Michael and Susan Dell Center for Healthy Living.
10. Analyzed year 2 data (pre-COVID-19).
11. Analyzed data collected during COVID-19 .

12. Continuing calibrating agent-based model, and developing a COVID-19 model.

13. Developed geospatial database in linkage with GPS and accelerometry data.

14. Developed manuscripts, three in print and two under review:

a) Published a paper in Public Health Nutrition titled "Validation of the FRESH Austin Food Frequency Questionnaire Using Multiple 24-hour Dietary Recalls"

b) Published a paper in Nutrients titled "Correlates of transitions in food insecurity status during the early stages of the COVID-19 pandemic among ethnically diverse households in Central Texas" utilizing COVID-19 specific data

c) Published a paper in International Journal of Environmental Research and Public Health titled "A Multi-Pronged Evaluation of a Healthy Food Access Initiative in Central Texas: Study Design, Methods, and Baseline Findings of the FRESH-Austin Evaluation Study" presenting the overall study design, methodology, and baseline survey findings

d) Submitted a manuscript to Public Health Nutrition titled "Examining geographic food access, food insecurity, and urbanicity among diverse, low-income participants in Austin, Texas" using baseline survey data.

e) Submitted a manuscript to International Journal of Behavioral Nutrition and Physical Activity titled "Who shops at their nearest grocery store? Exploring disparities in geographic food access among a low-income, racially/ethnically diverse cohort in Central Texas" using baseline survey data

2.5. Have any of the major goals/specific aims or milestones for the project changed since the award?*

Yes

If so, please list the goal(s) that have changed and provide justification for the change from the approved goals.*

Goals have slightly altered since the previous report, in which we described our study's expansion. We reported wanting to measure the effects of COVID-19 pandemic on participants' food purchasing and consumption behaviors and were able to do so via two surveys in 2020. In addition, we also collected qualitative data and conducted one-on-one interviews with 82 participants to further assess behaviors around food access and how they differed between those who were food insecure prior to COVID-19, those who were newly food insecure, and those who were consistently food secure.

Lastly, our original goal, and one that was reported last year, was to conduct a final survey of cohort participants in year 3. The third survey would have been similar to year 2 and baseline surveys to allow examination of time trends in food-related behaviors and environments. However, data collection of the year 3 survey could not happen due to the ongoing nature of the pandemic that interrupted implementation of the Fresh for Less intervention, and had dramatic changes in participant behaviors in real and perceived environments.

2.6. What was accomplished under the goals/specific aims or milestones for the project?*

Please describe in detail, 1) major activities; 2) specific objectives; 3) significant results, including major findings, developments, or conclusions (both positive and negative); and 4) key outcomes or other achievements. Include a discussion of stated goals not met. The emphasis in reporting in this section should focus on accomplishments. In the response, emphasize the significance of the findings to the scientific field. This section can be as technical as the PI would like.

One of the central aims of the study is collecting high quality data with sufficient sample size to yield reliable insights on food purchasing and consumption patterns in the Eastern Crescent of Austin, to inform our agent-based model of food purchasing behaviors. We accomplished surveying 400 residents in year 1, and resurveying 81% of the original cohort in year 2.

By year 3, the SARS-CoV-2 virus took over the world disrupting supply chains, distribution, and eating patterns. This interruption of the Fresh for Less initiative meant that our original goal of obtaining Year 3 data had to be abandoned. Nevertheless, we believe that our Year 1 and Year 2 survey data together include sufficient detail to support our conclusions regarding the impact of Fresh for Less. Moreover, Year

1 data is being used to parameterize an agent-based model, which continues to yield important insights into food purchase and consumption behavior under different scenarios.

While we dropped our originally planned Year 3 data collection, we decided to collect survey and interview data from the community at different stages of the pandemic during 2021, prior to the widespread availability of vaccines. We were at the forefront of measuring the impact this pandemic had on our cohort participants and we were able to document within-person changes in food security status, food consumption and eating behaviors from pre-pandemic, to early pandemic, to late pandemic among a same group of people. We found that food insecurity increased about 11% when the pandemic first started and decreased 9 months into the pandemic, but still not at pre-pandemic numbers:

[chart]

Food Insecurity (FI) Among Respondents, n=303

FI before COVID:

"never" = 66%

"sometimes or often" = 34%

FI start of COVID:

"never" = 55%

"sometimes or often" = 45%

FI 9 months into COVID:

"never" = 63%

"sometimes or often" = 37%

The findings from our COVID surveys allowed us to submit 4 abstracts to APHA (for 2021 American Public Health Association conference) and include results such as that participants (n=304) at start of COVID-19 pandemic were primarily concerned with risk of COVID-19 infection (81.5%), food access (60.4%), financial stability (59.7%), medical access (37%), and housing affordability (29%). Income was inversely associated with concerns relating to financial stability ($p<0.001$), food access ($p<0.001$), and housing affordability ($p=0.005$), and directly associated with concerns of coronavirus infection ($p<0.001$). White respondents had lower odds than Latinos of reporting concerns about housing (OR=0.5, 95% CI = 0.2-0.9), food access (OR=0.4, 95% CI = 0.2-0.7), and financial stability (OR=0.4, 95% CI=0.2-0.7). Respondents in households with children (versus without) were more concerned about food access ($p=0.10$).

We also found that 57.2% of the sample size reported declines in or maintenance of poor mental health. Hispanics were 2.53 (95% CI, 1.55-4.14) times more likely to have improved or maintained good mental health, compared to non-Hispanics. Most of the sociodemographic factors did not explain the differences in mental health status between Hispanics and non-Hispanics; however, adjusting for age and for children in household reduced these differences. Of the behavioral factors examined, change in sedentary time was independently associated with improvements in mental health but did not explain the differences between Hispanics and non-Hispanics.

In another abstract analysis, we restricted our sample to cohort members that completed the FRESH Y2, COVID-1 and COVID-2 surveys to see what socioeconomic factors were associated with experiencing food insecurity during the pandemic in any capacity. In the adjusted model, individuals with a household income less than $45,000 in 2019 and individuals who experienced difficulties accessing food during the pandemic were more likely to experience food insecurity during COVID-19 than those who were higher income in 2019 and did not experience issues with food access during COVID-19 respectively. These findings suggest that the COVID-19 pandemic did not create new food insecurity disparities. Rather, the pandemic exacerbated pre-existing disparities.

Our second goal was to collect qualitative data to obtain a more in-dept understanding of the quantitative data: fruit intake increased significantly during the pandemic, but not vegetable intake. We explored the relationship of FV intake by person’s food security status and whether those who were newly food secure vs. those who were food insecure prior to early 2020 had differing purchasing and consumption behaviors, as well as budgeting allocations. It was found that participants who reported eating more fruits were more likely to be in the food secure than food insecure groups, and the reason for eating more fruits was to stave off COVID-19 (by improving immune system). An interesting finding we discovered during the interviews was that people whom we categorized as food insecure, via the 2-item validated screener, do not see themselves as such. They stressed obtaining food from charitable organizations, such as food banks, and therefore do not consider themselves as having issues with getting enough food to eat.

In terms of our agent-based model, we have made substantial progress. When we started this work, the first version of the model included a simulated urban environment with randomly placed food stores and agents, and very simple rules to determine food purchasing behaviors. Namely, the first version of the model assumed that food purchasing behaviors were fully driven by distance to food outlets and available resources (money) for food. Following this first iteration, and informed by our primary data (survey, GPS) and secondary data sources, we have continued to improve on our model. The next iterations of the model were as follows: version 2 of the model assumed that agent's purchasing decisions were driven by distance to food outlets, resources (money for food), time for shopping, cooking and eating, and food preferences. In this second version of the model, food outlet locations, prices, agent income, and available time for food purchasing, preparation and intake were still being randomly assigned. A third iteration of the model also assumed decisions on food purchasing are a result of distance to food outlets, resources (money for food), time available for shopping, preparing and consuming food, and food preferences, but a major enhancement relative to version 2 was made: only income and time were being randomly assigned, whilst food outlet location and prices were based on actual primary data from our study. In this third version of the model, food preferences were also being defined by primary data from baseline, and were driven by race/ethnicity, as Hispanics tend to consume more vegetables than other groups. Version 3 of the model has since undergone several iterations of calibration for qualitative and quantitative accuracy.

One important decision that we made in this period was to use vegetable purchasing (versus fruit and vegetable purchasing) as the key outcome of interest for the model. This decision was made after observing, through multiple iterations of simulations to emulate the business as usual scenario in Austin, Texas, that the model was qualitatively working well, relative to our primary data findings, for predicting vegetable purchasing and intake, but not so much for fruit intake. This was an important finding and subsequent modeling decision, as it shows that the decision steps involved in determining vegetable versus fruit purchasing and intake behaviors are quite different. We ultimately opted for building a model that could capture the nature of vegetable purchasing and intake decisions, because vegetable intake is known to be lower than fruit intake in most populations, and is harder to change. In fact, there is evidence showing that given its sweetness, we are naturally "wired" to like fruits, while vegetable preferences are acquired – and thus are more challenging from the behavior change perspective.

In addition to being focused on vegetable purchasing and intake as the primary outcomes, our new model also integrates new rules. Namely, the importance of variety of items/services offered at a given food purchasing location as an important driver of the decision of where to shop for food. Also, we have developed a "COVID-19" version of the model, to understand the effect of COVID-19 transmission risk on food shopping behaviors and on food security. Finally, we have integrated the Fresh for Less exact locations as part of our simulated environment, and are currently in the final stages of the calibration process for the business as usual scenario. We are now at a stage in which we are defining final food policy environment scenarios to simulate. The types of questions we hope to answer with our model include: what is the minimum volume of Fresh for Less outlets that need to be implemented across the city for it to make a significant difference in population levels of vegetable purchasing and intake patterns? Do different Fresh for Less outlets (e.g., healthy corner stores, farm stands, or mobile markets) have different volume thresholds? Does placement by area of the city, or in relation to distance to other food vending stores, matter for reducing disparities? How did COVID-19 modify food access, vegetable food purchasing and intake, and food security among Austinites? We are working on a manuscript that we hope to submit by December 31 which will have more detailed information on modeling efforts and results.

Another activity taking place this period is the integration of our processed GPS data from baseline with accelerometry data (to draw connections between mode and routes of travel for food purchasing, and other health behaviors, like physical activity), and with Geographic Information Systems data. An analysis is underway to characterize the built environment of highly accessed versus less accessed food vending locations in Austin, to better understand the spatial determinants of food access in the area.

2.7. Discuss efforts taken to ensure the approach is scientifically rigorous. Include approaches taken to ensure robust and unbiased results.*

We have taken all the steps necessary when conducting scientific studies in an academic setting. Examples of these activities include: IRB approval was received; surveys were developed using best-practices and reviewed by experts and members of target audience group; data collectors were trained by experienced researchers; all data was collected using standardized protocols; and data were analyzed by experts in analytical methods.

2.8. List key stakeholders that could be served by results of this project.*

• City of Austin Office of Sustainability
• Non-profits working to improve healthy food access (i.e., SFC, Farmshare, Good Apple, Central Texas Food Bank, etc.)
• City of Austin City Council Members
• Travis County Officials
• City/County officials in similar rapidly growing communities that experience disparities in healthy food access and food security
• City of Austin Food Policy Board

2.9. How have the results of this reporting period been disseminated to communities of interest?*

Describe how the results have been disseminated to communities of interest. Include any outreach activities that have been undertaken to reach members of communities who are not usually aware of such activities, to enhance public understanding and increase interest in learning and careers in science, technology, and the humanities. Reporting the routine dissemination of information (e.g., websites, press releases) is not required. For awards not designed to disseminate information to the public or conduct similar outreach activities, a response is not required; the grantee should write “nothing to report.” A detailed response is only required for awards or award components that are designed to disseminate information to the public or conduct similar outreach activities. Note that scientific publications and sharing research sources will be reported under Information Products.

Results from the COVID-19 study have been shared with the City of Austin’s Office of Sustainability, as well as the board and staff of Episcopal Health Foundation. Presentations on the FRESH Austin study have also been given at conferences and webinars and shared with the other FFAR grant awardees. There are also two blogs written on COVID-19 impact on food purchasing and consumption, currently hosted on the Michael and Susan Dell Center for Healthy Living website. Additionally, our work has been shared with the scientific community via multiple conference presentations (APHA 2020, CFRAE January 2021, APHA 2021, CFRAE November 2021, and others), and the publication of 3 peer-reviewed manuscripts.

No results have been specifically disseminated to the public outside of these initiatives. However, we are working with Edwin Marty (Food Manager at the City of Austin’s Office of Sustainability to coordinate such activities in the future.

2.10. Describe challenges or delays encountered during project and actions that were taken to resolve them.*

Only describe significant challenges that may have impeded the research and emphasize their resolution.

Since the last reporting period, the challenge we ran into was duration of the pandemic and continuing uncertainty about its evolution. Our belief in early 2020 was that normal, pre-pandemic activities would resume by late 2020, and we would be able to conduct surveys continuing measuring strategies of the Fresh for Less Initiative. Initially there were 3 strategies: farm stands, mobile markets, corner stores. Last year we reported that farm stands and mobile markets merged into a single program and that the corner stores no longer were part of the Initiative. However, we did not anticipate that the one remaining strategy would go away in early 2020, and then become virtual in late 2020. Had we surveyed participants in year 3 with the same survey from years 1 and 2, the responses would not have been comparable across years; therefore, we were only able to analyze data from baseline to year 2. To better understand the pandemic, we expanded our scope of study and created a COVID survey which was distributed at two time-points. This allowed us to capture food purchasing and consumption behaviors early in the pandemic, and later in the pandemic with the same group of people.

2.11. What opportunities for training and professional development has the project provided during this reporting period?*

If the research is not intended to provide training and professional development during this period, state "Nothing to Report." For all projects reporting graduate student and/or post-doctoral participants, grantees are encouraged to describe how Individual Development Plans (IDPs) are used to help manage the training for those individuals.

During this reporting period, FRESH Austin team comprised of masters-level and doctoral-level students, one post-doctoral scholar, as well as one undergraduate Dell Health Scholar. Under the PI and Project Director's direct supervision, the undergraduate scholar and graduate level students were trained on survey creation, data cleaning, interview conduction, and qualitative data analyses. Training on statistical software was provided to staff and some graduate students, as well as guidance on the analytical portion of abstract submissions was provided by Dr. Ranjit, Co-I on this project. Dr. Salvo, Co-I on this project, has also provided direct mentorship to students and postdocs on topics ranging from analysis tasks to abstract and manuscript development. Dr. van den Berg provided opportunity and guidance to the doctoral level students for presentation of FRESH data. No Individual Development Plans were used.

All FRESH Austin team members were encouraged to attend national webinars on food insecurity, evaluation protocols, and COVID-19 effects on health.

2.12. Please indicate the number of undergraduate and graduate students, post-doctoral scholars, or other educational components involved during this reporting period. If other education components are involved, please describe them in detail.*

There were two master-level students, two doctoral students, one undergraduate scholar, and one post-doctoral fellow on the project during this reporting period.

3. INFORMATION PRODUCTS

3.1. Please list the type(s) of information products (e.g., scholarly publications, reports or monographs, workshop summaries or conference proceedings, video, audio, images, models, software, curricula, instruments or equipment, intervention, etc.) produced during this reporting period resulting directly from the FFAR award.*

3 published scholarly publication, two scholarly publication under review,

2 blog posts,

6 conference presentations,

4 recently submitted conference abstracts

1 webinars

1 GPS report.

Additionally, we have 3 manuscripts utilizing FRESH-Austin data that will be submitted to journals by December 31, 2021, and at least 4 more manuscripts that are in development and will be submitted in early 2022.

3.2. Please provide a list of citations for the information products produced during this period.*

1. Janda, K.M., Ranjit, N., Salvo, D., Nielsen, A., Lemoine, P., Casnovsky, J., & van den Berg, A. (2021). Correlates of transitions in food insecurity status during the early stages of the COVID-19 pandemic among ethnically diverse households in Central Texas. Nutrients, 13(8), 2597. https://doi.org/10.3390/nu13082597
2. Janda, K.M., Ranjit, N., Salvo, D., Nielsen, A., Akhavan, N., Diaz, M., Lemoine, P., Casnovsky, J., & van den Berg. A. (2021) A multi-pronged evaluation of a healthy food access initiative in central Texas: Study design, methods, and baseline results of the FRESH-Austin study. International Journal of Environmental Research and Public Health, 18(20). https://doi.org/10.3390/ijerph182010834
3. Jovanovic CE, Whitefield J, Hoelscher DM, Chen B, Ranjit N, van den Berg AE. Validation of the FRESH Austin Food Frequency Questionnaire Using Multiple 24-hour Dietary Recalls. Public Health Nutrition. 2021 May 26:1-26.
4. Janda, K.M., Salvo, D., Ranjit, N., Hoelscher, D.M., Nielsen, A., Lemoine, P., Casnovsky, J., & van den Berg, A. (Under Review – International Journal of Behavioral Nutrition and Physical Activity) Who shops at their nearest grocery store? Exploring disparities in food access among a racially/ethnically diverse cohort in Central Texas.
5. Janda, K.M., Ranjit, N., Salvo, D., Hoelscher, D., Nielsen, A., Casnovsky, J., & van den Berg, A. (Under

Review – Public Health Nutrition) Examining Geographic Food Access, Food Insecurity and Urbanicity among Diverse, Low-Income Participants in Austin, Texas.
6. Zhang, Y., 2021. Concerns among Central Texas Residents during COVID-19 Pandemic. [Blog] Michael and Susan Dell Center for Healthy Living, Available at: https://sph.uth.edu/research/centers/dell/blog/posting.htm?s=Concerns-among-Central-Texas-Residents-during-COVID-19-Pandemic
7. Salhab, R., Herrera, V., 2020. Snapshot of Concerns among Texas Residents during the COVID-19 Pandemic. [Blog] Michael and Susan Dell Center for Healthy Living, Available at: https://sph.uth.edu/research/centers/dell/blog/posting.htm?s=Snapshot-of-Concerns-among-Texas-Residents-during-the-COVID-19-Pandemic
8. Janda, K., Ranjit, N., Salvo, D., Nielsen, A., Lemoine, P., Casnovsky, J., & van den Berg, A. Correlates of food insecurity transitions during early COVID-19 in Central Texas. Food Insecurity, Neighborhood Food Environment, and Nutrition Health Disparities: State of the Science Workshop– Food Insecurity Track. Virtual. September 2021.
9. Janda, K., Salvo, D., Ranjit, N., Nielsen, A., Hoelscher, D., Casnovsky, J., & van den Berg, A. Who shops at their nearest grocery store? Exploring disparities in food access. Food Insecurity, Neighborhood Food Environment, and Nutrition Health Disparities: State of the Science Workshop – Neighborhood Food Environment Track. Virtual. September 2021.
10. Nielsen, A., Salvo, D., Janda, K., Ranjit, N., Zhang, Y., van den Berg, S. Sociodemographic factors associated with economic-, food security- and health-related concerns during COVID-19 among adults from low-income, majority-minority urban areas. APHA’s 2021 Annual Meeting and Expo (Oct. 24-27). American Public Health Association.
11. Nielsen, A., Ranjit, N., Janda, K., Jovanovic, C., Hood, R., Akhavan, N., ... & van den Berg, A. (2020, October). Baseline data of the fresh Austin study: Use of nontraditional food retail outlets by low-income communities in Austin. In APHA's 2020 VIRTUAL Annual Meeting and Expo (Oct. 24-28). American Public Health Association.
12. Jovanovic, C., Ranjit, N., Nielsen, A., Hood, R., Flores-Thorpe, S., Janda, K., ... & van den Berg, A. (2020, October). Associations of purchasing behavior, consumption of fruits and vegetables (FV) and scratch cooking among a diverse population in Austin, TX. In APHA's 2020 VIRTUAL Annual Meeting and Expo (Oct. 24-28). American Public Health Association.
13. Janda, K., Ranjit, N., Salvo, D., Nielsen, A., Casnovsky, J., van den Berg, A. Predictors of Food Insecurity during the COVID-19 pandemic in a low-income cohort in Central Texas. Submitted in APHA’s 2021 Annual Meeting and Expo (Oct. 24-27). American Public Health Association.
14. van den Berg A., Janda K., Salvo D., Nielsen A., and Ranjit N. Impact of a mobile market intervention on food security and fresh fruit and vegetable purchase and consumption among a low-income, ethnically and racially diverse population. Submitted to International Society of Behavioral Nutrition and

Physical Activity Annual Meeting 2022. (Under Review).

15. Salvo, D., Garcia, L., Reis, R., Goel, R., Schiperijn, J., Hallal, P., Ding, D., Pratt, M., Lemoine, P., Janda, K., Ranjit, N., Nielsen, A., and van den Berg, A. Examining the impact of large-scale built and food environmental changes on physical activity and healthy eating using agent-based modeling. Submitted to International Society of Behavioral Nutrition and Physical Activity Annual Meeting 2022. (Under Review).

16. Salvo, D., van den Berg, A., Jauregui, A., Janda, K., Lanza, K., Hoelscher, D., and Villa, U. Integrating Geographic Positioning Systems and accelerometer monitor data for assessing the spatiotemporal patterns of health behavior. Submitted to International Society of Behavioral Nutrition and Physical Activity Annual Meeting 2022. (Under Review).

17. Janda, K., Salvo, D., Ranjit, N., Nielsen, A., Lemoine, P., and van den Berg, A., Who shops at their nearest grocery store? A cross-sectional exploration of disparities in geographic food access among a low-income, racially/ethnically diverse cohort in Central Texas. Submitted to International Society of Behavioral Nutrition and Physical Activity Annual Meeting 2022. (Under Review).

18. Webinar: Janda, K., Nielsen, A., Zhang, Y., Viri, O., Salvo, D., van den Berg, A. (2021, March 22). Exploring healthy food access and consumption among low-income central Texas residents during COVID-19: the FRESH Austin Study. Michael and Susan Dell Center for Healthy Living.
https://sph.uth.edu/research/centers/dell/webinars/webinar.htm?id=2bb89e19-7fef-46c0-aea0-b95ee3146945

19. Webinar: Gundersen, C., van den Berg, S. (2020, June 30). Increases in Food Insecurity due to COVID-19: What Can be Done? Michael and Susan Dell Center for Healthy Living.
https://sph.uth.edu/research/centers/dell/webinars/webinar.htm?id=7e7128e1-9c5f-461a-a80d-c75ac39411b7

20. Salvo, D. (2020). FRESH Austin Study: Baseline GPS Data Analysis Report. Unpublished.

3.3. Are there publications or manuscripts accepted for publication in a journal or other publication (e.g., book, one-time publication and monograph) during the reporting period resulting directly from the FFAR award?*

Yes

If yes, please provide citation.

1. Jovanovic CE, Whitefield J, Hoelscher DM, Chen B, Ranjit N, van den Berg AE. Validation of the FRESH Austin Food Frequency Questionnaire Using Multiple 24-hour Dietary Recalls. Public Health Nutrition. 2021 May 26:1-26.
2. Janda, K.M., Ranjit, N., Salvo, D., Nielsen, A., Lemoine, P., Casnovsky, J., & van den Berg, A. (2021). Correlates of transitions in food insecurity status during the early stages of the COVID-19 pandemic among ethnically diverse households in Central Texas. Nutrients, 13(8), 2597. https://doi.org/10.3390/nu13082597
3. Janda, K.M., Ranjit, N., Salvo, D., Nielsen, A., Akhavan, N., Diaz, M., Lemoine, P., Casnovsky, J., & van den Berg. A. (2021) A multi-pronged evaluation of a healthy food access initiative in central Texas: Study design, methods, and baseline results of the FRESH-Austin study. International Journal of Environmental Research and Public Health, 18(20). https://doi.org/10.3390/ijerph182010834
4. Janda, K.M., Salvo, D., Ranjit, N., Hoelscher, D.M., Nielsen, A., Lemoine, P., Casnovsky, J., & van den Berg, A. (Under Review – International Journal of Behavioral Nutrition and Physical Activity) Who shops at their nearest grocery store? Exploring disparities in food access among a racially/ethnically diverse cohort in Central Texas.
5. Janda, K.M., Ranjit, N., Salvo, D., Hoelscher, D., Nielsen, A., Casnovsky, J., & van den Berg, A. (Under Review – Public Health Nutrition) Examining Geographic Food Access, Food Insecurity and Urbanicity among Diverse, Low-Income Participants in Austin, Texas.

At least 7 more manuscripts in development, with 3 planning to be submitted by December 31, 2021 and 4 planning to be submitted in early 2022.

3.4. Website(s). List the URL for any internet site(s) that disseminates the research activities. A short description of each site should be provided. It is not necessary to include the publications already specified above.*

This project is being conducted at the University of Texas Health Science Center at Houston School of Public Health in Austin. Description of the project can be found here: https://sph.uth.edu/research/centers/dell/project.htm?project=9707359b-4d0d-426e-b915-760809f89b4a

3.5. Have inventions, patent applications, and/or licenses resulted from the award during this reporting period?*

No

3.6. Are any of the information products produced during this reporting period confidential, proprietary, or subject to special license agreements?*

No

3.7. Beyond depositing information products in a repository, what other activities have you undertaken to ensure that others (e.g., researchers, decision makers, and the public) can easily discover and access the listed information products? What other activities have you undertaken to ensure that other can access and re-use these data in the future?*

We have a project webpage, housed on Michael and Susan Dell Center for Healthy Living's website, with general information on the study including PI's and Co-I's contact information. The project webpage will be updated with links to publications as they become available.

We have also presented some of our findings at conferences, for example APHA in 2020 and 2021, as well as hosted a webinar, presented to the City of Austin and fellow FFAR grant recipients.

We have published one paper on the validation of the FFQ, submitted a manuscript examining food insecurity transitions during early stages of COVID-19, and are working on writing other papers (study methodology, methodology for cleaning and processing GPS data for conducting spatial analysis of food purchasing behaviors, barriers and facilitators of F&V consumption across racial/ethnic groups, etc.).

4. DATA MANAGEMENT

4.1. During this period, did the project generate any data?*

Data generation includes transformation of existing data sets and data from existing resources (e.g., maps and imageries).

Yes

Please list the data generated in this reporting period.

Briefly, for the complete project we have quantitative cohort data collected at baseline and year 2; we have quantitative data for COVID Survey as time 1 (Spring 2020) and time 2 (Fall 2020); we have qualitative COVID data from interviews conducted in Spring 2021: GPS data from baseline; accelerometry and GPS data from baseline; store audits from baseline; built environments assessments at baseline. Details of all data (metadata) obtained are available in the Texas Data Repository.

All individual quantitative survey data has been processed into Stata files and recoded. Quantitative data will be placed into the Repository as soon as cleaning is completed.

Qualitative interview data have been transcribed, and themes analyzed. A report on the interview data will be placed in the repository on completion. Baseline GPS data from 100 participants have been processed, and the report is available on the repository.

4.2. If you list multiple data sets, are these data sets related?*

Yes

If so, please provide a short description of how they are related.*

All of our individual level datasets are linked by participant ID. Baseline and year 2 data sets are linked to the COVID surveys by participant ID as well. Device-based geo-positional systems (GPS) data were from a subset of the 400 participants (n=100) and are linked via participant ID number as well.

Resident address data can be matched with geocoded data on stores and built environment assets.

4.3. Please provide copies of relevant metadata records to support FFAR's mission of enhancing the discoverability of FFAR funded project data and information products. Upload copies of records and a simple file inventory, if necessary, in a compressed folder.*

FFAR is developing a more nuanced Data Management Policy. As part of this effort, we are introducing new questions into the Annual Reports on Data. All questions on this page are required to have a response. In cases where your answer may be repeated, you are welcome to copy-paste relevant information. Thank you for your patience as we update our reporting.

Are the data experimental, observational, simulation, or compiled/derived? Check all categories that apply.

Responses Selected:

Observational: Data captured through observation of behavior or actions of the situation as is (e.g., surveys, sensors, monitors)
Simulation: Data generated by developing and applying a computer model that simulates system behavior (e.g., ecosystem, behavioral)

For each checked box, please provide a brief description of the data including where the data are collected / sourced.

Observational data included field data collected by completing store audits and built environment observations. Additional observational data included survey data were collected two times during this reporting period. Surveys were administered in July/July 2020, as well as November/December 2021. These data were collected with the purpose of testing an intervention (the Fresh for Less Initiative). Additional surveys were collected in 2021 to assess COVID-19 impacts on food security and food purchase behaviors in this population.

Simulation: agent-based model has been revised and has undergone several iterations of calibration for qualitative and quantitative accuracy. Final data can be provided by end of the year (2021)

Describe (what, where, when) and list additional data (not captured in question above), plus samples, physical collections, software, curriculum materials and other materials produced during this period. Who is the expert or person(s) tasked with data collection (please provide their name and affiliation with the project)? This can be any member of the team. For data-focused projects, that individual is expected to be key personnel.

Additional data include interview data (n= 82) collected via phone interviews by trained team members.

The person overseeing data collection was Dr. van den Berg, PI of the FRESH Austin study.

The PI, two Co-Is, Post-Doctoral Fellow and the Project Director worked with a consultant, Pablo Lemoine, on the agent-based model creation. Pablo Lemoine is in charge of developing agent-based models and is responsible for all data arising from those simulations.

Provide how the data are organized and managed for all data created and utilized during this project. This could include the relevant data standardization, data formats, data definitions (e.g., data dictionary, ontologies), data organization, and/or where the data is stored. Please note if any of these data are confidential, and if so, when you would anticipate being able to share them publicly in the future.

All data collected will be stored on a secured server from The University of Texas Health Science Center at Houston. Data collected will be deidentified and uploaded to the Texas Data Repository by August 2024 while all materials (instruments and protocols) were uploaded mid-December 2021. Principal Investigator's contact information will be added to the repository should anyone wish to see the data prior to year 2024.

FFAR supports the FAIR principles. We would like to understand how your project aligns with FAIR principles (for definition, see Go Fair, for more information see Nature Magazine). Does your data align with the 4 principles in FAIR? Please check all boxes that apply and provide a brief description:

Responses Selected:

Findable – how could your data be found (e.g., online searches, publications)?
Access - how will the data be accessed and shared by researchers and the general public, respectively (e.g., is the data publicly accessible, proprietary, temporarily proprietary)?

For each checked box, please provide a brief description.

Findable: Data will be findable three years after study’s completion and will be placed in the Texas Data Repository in 2024. Texas Data Repository (https://dataverse.tdl.org/) is built in an open-source application called the Dataverse software, developed and used by Harvard University. Survey instruments, protocols, and published papers, however, are currently available: https://data.tdl.org/dataverse/fresh-austin

Access: All survey instruments, protocols, and published manuscripts that are open-access (as they come out in print) are placed in the FRESH Austin folder within the Texas Data Repository. Data will be deidentified and uploaded three years after study’s completion, but other project materials will be available sooner.

Interoperable: At the moment, no attempt has been made to connect data to other sources.

Reusability: Instructions for use of the data by non-UTHealth researchers are currently being developed. In addition, any publications (or links) developed from these data will be uploaded into the Texas Data Repository for use as resources by other researchers.

What are the provisions for the data for the appropriate protection of privacy, confidentiality, security, intellectual property, or other rights or requirements?

As of now, all data is maintained on the UTHealth servers, which are fully secured and password protected (two-factor authentication). Data uploaded to Texas Data Repository will fully be deidentified, including stripped of any geographical location data.

Describe the plans for archiving data, samples, and other research products, and for preservation of access to them. Refer to the FFAR Data Guidance Document for more information.

Original versions of the data will be maintained on UTHealth servers for at least seven years following the completion of the study. Any hard copies of the data will be maintained under lock and key in the PI's office for at least seven years following study's completion.

Will this project coordinate with existing FFAR-supported grants and their data products? If so, provide a brief description of how, including the project title and PI of the existing FFAR work If not, provide a brief rationale for why this project does not coordinate with existing FFAR-supported work.

The Austin team is working with four other FFAR funded groups to continue integration of results and lessons learner. Several workshops are scheduled and we intent to meet in person in DC during late Spring.

Please provide any feedback on these new data questions including any areas where you would like to see more clarity or explanation.

We appreciated the questions asked.

5. CERTIFICATION

The undersigned hereby certifies that to the best of their knowledge and belief, the above report and all supporting documents are true, accurate, and complete. I am aware there is a significant penalty for submitting false or misleading information.

By signing below, I agree that my electronic signature is legally binding, equivalent to, and has the same validity and meaning as my handwritten signature. I will not claim otherwise.

PI Full Name*

Cory Leahy for Ms. Joy Casnovsky

PI Electronic Signature*

Date*

12/17/2021

Authorized Signing Official (ASO)*

Mr. Mark Bethell

ASO Electronic Signature*

Date*

12/17/2021

Final Invention Report and Certification

Completed - Dec 17 2021

Final Invention Report and Certification

FINAL INVENTION REPORT AND CERTIFICATION

A Final Invention Report must be submitted within 90 days after the expiration or termination of a FFAR funded research grant. The Report must include all inventions which were conceived or first actually reduced to practice during work performed under the funded grant, from the original effective start date of the grant through the date of completion or termination. The Final Invention Report and Certification in no way relieve the principal investigator, project director, or the grantee, of the obligation to assure that all inventions are promptly and fully reported directly to the Foundation for Food and Agriculture Research (FFAR), as required by terms of the grant or award.

If no inventions were conceived or reduced to practice during the funded research, write a statement to that effect in the first block under item Title of Invention. The Authorized Organization Representative (AOR) must certify the Report prior to submitting it.

Grantee:*

Sustainable Food Center, Inc.

Grant ID:*

CA18-SS-0000000050

Project Title:*

Evaluating food access strategies in Austin to improve healthy food consumption and food security

Date of the Report:*

12/17/21

INVENTION(S)

If no inventions were conceived or reduced to practice during the funded research, write “NONE” in the space provided below.

	Name of Inventor	Title of Invention	Date Reported to FFAR
1	NONE		
2			
3			
4			
5			
6			
7			
8			
9			
10			

The undersigned hereby certifies that to the best of their knowledge and belief, the information in the Report below are true, accurate, and complete. You also certify that the Report lists all inventions which were conceived and/or first actually reduced to practice during the work under the above-referenced FFAR grant or award for the period of:

Project Start Date:*

02/01/2018

Project End Date:*

09/30/2021

Authorized Organization Representative (AOR) Name:*

Mr. Mark Bethell (signed by Cory Leahy)

AOR Title:*

Executive Director

AOR Signature:*

Please use your cursor to sign on the line below:

Date:*

12/17/2021

Final Financial Status Report

Completed - Dec 17 2021

FFAR Final Financial Report - 3 yrs

Filename: FFAR_Final_Financial_Report_-_3_yrs.xlsx **Size:** 29.4 kB

Final Equipment Inventory Report

Completed - Dec 17 2021

Equipment-Inventory-FINAL

Filename: Equipment-Inventory-FINAL.pdf **Size:** 1.1 MB

Copies of Relevant Metadata Records

Completed - Dec 17 2021

Final Metadata Information

Filename: Final_Metadata_Information.docx **Size:** 21.2 kB

Other Modifications

Incomplete

No Cost Extension Request

Incomplete

No Cost Extension Request

NO COST EXTENSION REQUEST

This document describes the procedures that must be followed when a grantee determines that their activities will not be completed during the approved project timeline. A no-cost extension requires prior approval from the Director of Grants Management.

FFAR grants and cooperative agreements are awarded under a project period system. A project may be approved for multiple years (usually 3-5 years), and is generally funded in annual increments known as budget periods. With rare exceptions, budget periods are 12 months in duration. FFAR expects that grantees will complete all requirements of an award by the project period end date; however, a one-time no cost extension may be requested if the action is not prohibited in the grant agreement. The Director of Grants Management can approve up to 6 additional months but no additional funds are awarded to complete the funded tasks.

Note that a No Cost Extension Request should not be submitted for the sole purpose of expending remaining funds – such request will be disapproved. No Cost Extensions cannot be processed under expanded authority if the award project period end date has expired.

FFAR Notification

To ensure timely processing of a revised award action and orderly accomplishment of activities, a no-cost extension should be requested at least 45 days prior to the end of the project period by sending a request on official grantee letterhead that includes the following:

- Date
- Principal Investigator or Project Director name, Project Title, and Grant ID
- Point of contact – name, phone number, and email address
- Amount of additional time requested
- Reason(s) project could not be completed
- Description of the activities that will be completed during the proposed extension
- Timeline for completion of proposed activities, including time necessary to close-out the award and submit all final requirements to FFAR
- For late requests, a justification for missing the deadline
- Explain the effect a denial of the request will have on the project
- Two signatures – Authorized Signing Official and Project Director/Principal Investigator

NO COST EXTENSION REQUEST

Date:*

(No response)

Principal Investigator or Project Director name:*

(No response)

Project Title*

(No response)

Grant ID*

(No response)

Point of contact*

Name	(No response)
Phone Number	(No response)
Email Address	(No response)
Street Address	(No response)
City	(No response)
State	(No response)
Zip Code	(No response)

Amount of additional time requested:*

(No response)

Reason(s) project could not be completed:*

(No response)

Description of the activities that will be completed during the proposed extension:*

(No response)

Timeline for completion of proposed activities, including time necessary to close-out the award and submit all final requirements to FFAR*

(No response)

For late requests, a justification for missing the deadline

(No response)

Explain the effect a denial of the request will have on the project*

(No response)

Authorized Signing Official:*

Please use your cursor to sign on the line below:

Date Signed:*

(No response)

Project Director/Principal Investigator:*

Please use your cursor to sign on the line below:

Date Signed:*

(No response)

Optional - Additional Material Upload

Completed - Dec 17 2021

Final Report Tables - Question 2

Filename: Final_Report_Tables_-_Question_2.3_UsgV80Z.pdf **Size:** 246.5 kB

Review Round Final: Scientific Progress Report for: John Reich

In Progress - Last edited: Dec 20 2021
Score:

Review Task Form

Scientific Report Review

Award Information*

Grantee Name	Sustainable Food Center, Inc.
PI Name	Ms. Joy Casnovsky
Grant ID	CA18-SS-0000000050
Project Title	(No response)
Reporting Period	MM/DD/YYYY - MM/DD/YYYY
Award Program	Tipping Points

Spending Notes

This includes the approved project budget and any carryover from previous years.

FFAR Approved Budget (FFAR Funds and Match)	$2,197,581.62
Reported Expenditures	$2,184,742.34
Variance	$12,839.28

Project Summary*

Briefly summarize what the project tried to accomplish in this reporting period

(No response)

Did the PI accomplish the Goals and Objectives set for this reporting period?*

If the PI met SOME of the Goals and Objectives, select both choices. If the PI met ALL the Goals and Objectives, select 'Yes.' If the PI met NONE of the Goals and Objectives, select 'No.'

Responses Selected:

Yes
No

Please explain what was accomplished in this project*

(No response)

What is the SPD's overall evaluation of what the project was able to accomplish?*

Please provide a brief statement on the Return on Investment for FFAR and the public for funding this project.

(No response)

Final Scientific Progress Report Review Rating*

(No response)

Equipment Inventory Report

The Equipment Inventory Report is due 90 days after the project period end date. A complete inventory must be submitted for all major equipment acquired or furnished under this project with a unit acquisition cost of $5,000 or more. The inventory list must include the description of the item, manufacturer serial and/or identification number, acquisition date and cost, percentage of FFAR funds used in the acquisition of the item.

Equipment with a unit acquisition cost of less than $5,000 that is no longer to be used in projects or programs currently or previously funded by the FFAR may be retained, sold, or otherwise disposed of, with no further obligation to FFAR. If no equipment was acquired under this award, please write a statement to that effect.

FOUNDATION FOR FOOD & AGRICULTURE RESEARCH

Grantee Name: Sustainable Food Center
Grant ID: CA18-SS-0000000050
Grant Period: 2/1/18-9/30/21

Project Title: Evaluating food access strategies in Austin to improve healthy food consumption and food security
Principal Investigator: Joy Casnovsky
Report Date: Dec. 17, 2021

Item Description	Manufacturer	Serial Number	Quantity	Condition	Location1	Purchase Cost	% of FFAR funds to purchase	Date Received
No equipment was acquired under this award.								
Item	Manufacturer	#	#	Choose an item.	enter text	$	%	date
Item	Manufacturer	#	#	Choose an item.	enter text	$	%	date
Item	Manufacturer	#	#	Choose an item.	enter text	$	%	date
Item	Manufacturer	#	#	Choose an item.	enter text	$	%	date
Item	Manufacturer	#	#	Choose an item.	enter text	$	%	date
Item	Manufacturer	#	#	Choose an item.	enter text	$	%	date
Item	Manufacturer	#	#	Choose an item.	enter text	$	%	date
Item	Manufacturer	#	#	Choose an item.	enter text	$	%	date

1Location: complete physical address.

FOUNDATION FOR FOOD & AGRICULTURE RESEARCH

Certification

The undersigned hereby certifies that to the best of their knowledge and belief, the above report and all supporting documents are true, accurate, and complete. I am aware there is a significant penalty for submitting false or misleading information. By checking the box below, I agree that my electronic signature is legally binding, equivalent to, and has the same validity and meaning as my handwritten signature. I will not claim otherwise.

PI Full Name: Joy Casnovsky

Authorized Signing Official (ASO): Mark Bethell

☒PI Electronic Signature: Cory Leahy for Joy Casnovsky Date: 12/17/21

X ASO Electronic Signature: Mark Bethell Date: 12/17/21

Figure1 – Increasing locations and financial discounts

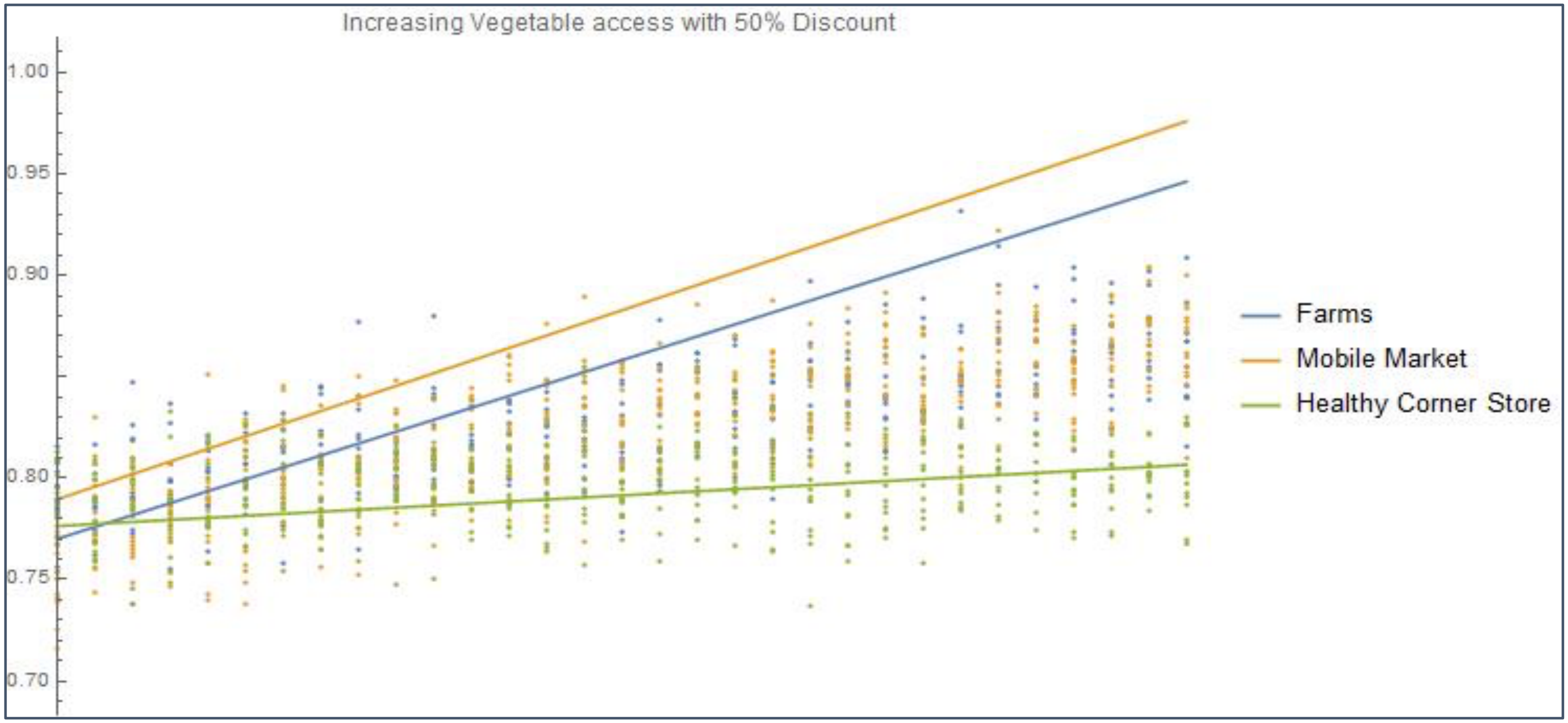

Table 1. Descriptive statistics of sociodemographic characteristics stratified by recruitment strategy among adults at baseline (FRESH Evaluation study Austin, TX, USA, October 2018–March 2019).

Variable	Full Sample	Confirmed Users	Geographically Exposed	Comparison	*p*-Value/ANOVA
	***n* = 400** % (N)/Mean (SD)	***n* = 130** % (N)/Mean (SD)	***n* = 185** % (N)/Mean (SD)	***n* = 85** % (N)/Mean (SD)	
Gender					
Female	70.5% (282)	73.85% (96)	68.11% (126)	70.59% (60)	
Male	29.25% (117)	26.15% (34)	31.35% (58)	29.41% (25)	0.69
Race/Ethnicity					
Hispanic/Latino	54.41% (216)	60.77% (79)	54.10% (99)	45.24% (38)	
Black	10.08% (40)	6.15% (8)	10.38% (19)	15.48% (13)	
White/Other	35.52% (141)	33.08% (43)	35.52% (65)	39.29% (33)	0.12
Household Income in 2017					
Under $25,000	23.04% (88)	32.81% (42)	16.09% (28)	22.50% (18)	
$25,001–$45,000	29.58% (113)	35.94% (46)	27.59% (48)	23.75% (19)	
$45,001–$65,000	18.32% (70)	17.19% (22)	18.97% (33)	18.75% (15)	
Over $65,000	29.06% (111)	14.06% (18)	37.36% (65)	35.00% (28)	**<0.01**
Food Assistance *					
Utilized Food Bank in Last Year	12.00% (48)	18.46% (24)	9.73% (18)	7.06% (6)	**0.02**
Utilized Free and Reduced Lunch in Last Year	26.50% (106)	36.92% (48)	24.86% (46)	14.12% (12)	**<0.01**
Utilized SNAP in Last Year	17.50% (70)	26.92% (35)	13.51% (25)	11.76% (10)	**<0.01**
Utilized Women, Infants, and Children (WIC) in Last Year	9.25% (37)	13.08% (17)	8.11% (15)	5.88% (5)	0.16
Food Insecurity					
Sometimes or Often	39.60% (158)	49.61% (64)	37.30% (69)	29.41% (25)	
Never	60.40% (241)	50.39% (65)	62.70% (116)	70.59% (60)	**<0.01**

* Could select more than one answer option., **bold** signifies significant differences at $p < 0.05$

Table 2. Fruit and vegetable purchasing, food shopping behaviors and motivations among adults at baseline (FRESH Evaluation study Austin, Texas, (October 2018–March 2019)).

Variable	Full Sample ***n* = 400** % (N)/Mean (SD)	Confirmed Users ***n* = 130** % (N)/Mean (SD)	Geographically Exposed ***n* = 185** % (N)/Mean (SD)	Comparison ***n* = 85** % (N)/Mean (SD)	***p*-Value/ANOVA**
Fresh Fruit and Vegetable Purchasing					
Fruit Pounds/Capita/Week	3.36 [2.97]	3.59 [3.40]	3.26[2.84]	3.22 [2.53]	$F(2,2) = 0.59$, $p = 0.56$
Vegetable Pounds/Capita/Week	4.65 [3.93]	5.03 [4.35]	4.32 [3.86]	4.80 [3.37]	$F(2,2) = 1.34$, $p = 0.26$
Shopped at Fresh For Less Location(s) *					
Shopped at Farm Stand	12.75% (51)	37.69% (49)	1.08% (2)	0.00% (0)	**<0.01**
Shopped at Mobile Market	15.00% (60)	40.77% (53)	3.78% (7)	0.00% (0)	**<0.01**
Shopped at Health Corner Store	5.00% (20)	12.31% (16)	2.16% (4)	0.00% (0)	**<0.01**
Most Important Factor When Deciding Where to Shop for Food					
Quality of Food	52.63% (210)	46.34% (57)	55.19% (101)	62.65% (52)	
Cost	25.96% (101)	27.64% (34)	24.04% (44)	27.71% (23)	
Variety of Food	12.34% (48)	15.45% (19)	13.11% (24)	6.02% (5)	
Quality of Store	4.88% (19)	4.88% (6)	5.46% (10)	3.61% (3)	
Cultural Variety	2.83% (11)	5.69% (7)	2.19% (4)	0.00% (0)	0.1

* Could select more than one answer option, **bold** signifies significant differences at $p < 0.05$

Table 3: Changes baseline – year 2 adjusted for Race/Ethnicity and Income

	Confirmed Users		Geographically Exposed		Comparison Group	
	Contrast (SE)	*p*-value	Contrast (SE)	*p*-value	Contrast (SE)	*p*-value
FV Consumption						
Fruit Consumption (Cups/Day)	0.01 (0.09)	0.92	0.005 (0.08)	0.95	0.21 (0.12)	0.07
Vegetable Consumption (Cups/Day)	0.03 (0.09)	0.72	-0.01 (0.08)	0.87	-0.08 (0.12)	0.52
FV Purchasing (lbs per capita/day)						
Fruit Purchasing (lbs per capita/day)	0.68 (0.37)	0.06	**0.73 (0.31)**	**0.02**	0.51 (0.46)	0.27
Vegetable Purchasing (lbs per capita/day)	0.08 (0.34)	0.82	-0.24 (0.29)	0.4	-0.21 (0.43)	0.63
Food Insecurity						
Sometimes or Often Food Insecure	**-0.10 (0.04)**	**0.01**	**-0.10 (0.03)**	**<0.01**	-0.05 (0.05)	0.36
Utilized Mobile Markets/FFL						
Shopped at Mobile Markets/FFL	0.09 (0.04)	***0.02***	0.04 (0.03)	*0.24*	0.06 (0.05)	*0.18*

Table 4. Demographic characteristics of the sample stratified by food insecurity transition status.

	Stayed Food Secure *n* = 163	Newly Food Insecure *n* = 66	Consistently Food Insecure *n* = 125	Total *n* = 354
	% (*n*)	**% (*n*)**	**% (*n*)**	**% (*n*)**
Race/ethnicity				
Latino and/or Black	41.83% (64)	71.43% (45)	90.08% (109)	64.69% (218)
Non-Hispanic White	58.17% (89)	28.57% (18)	9.92% (12)	35.31% (119)
Presence of children in the household				
No children in household	56.88% (91)	30.30% (20)	21.14% (26)	39.26% (137)
One or more child in household	43.12% (69)	69.70% (46)	78.86% (97)	60.74% (212)
Change in employment status or wages during the COVID-19 pandemic				
No change in employment status or wages during the COVID-19 pandemic	74.03% (114)	39.68% (25)	39.34% (48)	55.16% (187)
Lost job or wages during the COVID-19 pandemic	25.97% (40)	60.32% (38)	60.66% (74)	44.84% (152)
Language spoken at home				
Mainly Spanish spoken at home	19.75% (31)	41.94% (26)	71.90% (87)	42.35% (144)
Mainly English or other language spoken at home	80.25% (126)	58.06% (36)	28.10% (34)	57.65% (196)
Changes in shopping location during the COVID-19 pandemic due to Issues with...				
Availability (usual store did not have the food that was needed), affordability (other stores were cheaper than usual store), and/or accessibility (other stores were closer than usual store)	26.99% (44)	59.09% (39)	59.20% (74)	44.35% (157)
Did not report changing shopping locations due to issues with availability, affordability, and/or accessibility	73.01 (119)	40.91% (27)	40.80% (51)	55.65% (197)
Food availability issues during the COVID-19 pandemic				
Had little to no difficulties finding their normally purchased groceries during the COVID-19 pandemic	64.42% (105)	31.82% (21)	34.40% (43)	47.74% (169)
Sometimes or always had difficulties finding their normally purchased groceries during the COVID-19 pandemic	35.58% (58)	68.18% (45)	65.60% (82)	52.26% (185)
perceptions of food prices during the COVID-19 pandemic				
Food prices increased during the COVID-19 pandemic	61.35% (100)	71.21% (47)	77.60% (97)	68.93% (244)
Food prices decreased or stayed the same during the COVID-19 pandemic	38.65% (63)	28.79% (19)	22.40% (28)	31.07% (110)